

Automatic Screen Lock Program

for Privacy & Security

Using Teachable Machine AI + Python

EF1003 — Data Literacy Foundations(01)

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01 Idea & Motivation

The Problem

Leaving a computer unattended can expose sensitive information to anyone nearby.

Manual locking is often forgotten — especially in busy environments like libraries, offices, and labs.

Our Solution

A real-time webcam monitor that detects whether a person is present at the desk.

If the seat is empty for 5 consecutive seconds, Windows screen is automatically locked.

Practical, lightweight, and immediately usable in everyday situations.

02 Data Collection & Curation

370

person_existed images

136

no_person images

506

total training samples

Collection Strategy

- person_existed (370): 3 variations — solo Person A, solo Person B, both persons together
- no_person (136): empty chair/desk including various locations with various backgrounds and lighting.
- All images captured via webcam.

Curation & Quality Control

- Removed blurry, low-resolution, and poorly lit images
- Class imbalance noted (370 vs 136) — compensated by diversifying person_existed with 3 sub-types
- Ensured variety in angles and distances.

03 AI Model Training

Model & Hyperparameters

Platform: Google Teachable Machine

Base model: MobileNet (Transfer Learning)

Epochs: 50 — full passes over the dataset

Batch Size: 16 — images per gradient update

Learning Rate: 0.001 — standard default value

Default settings were kept as model performance was already optimal.

Performance Results

person_existed

100%

56 test samples

no_person

100%

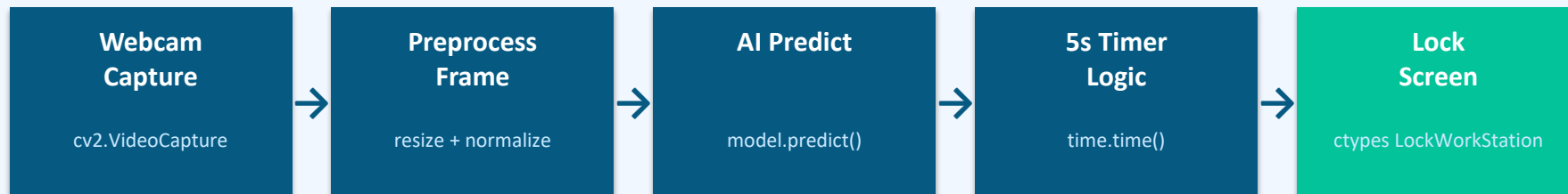
21 test samples

Discussion: Why near-perfect accuracy?

The two classes are visually very distinct. (Unlike analyzing complex poses or distinguishing between objects, it only checks whether there are objects or not.)

Potential failures: extreme lighting, partial occlusion, user not facing camera.

04 Program Implementation



Key Functions

<code>load_model_and_labels()</code>	Loads keras_model.h5 and parses labels.txt into a list
<code>preprocess_frame(frame)</code>	Resizes to 224×224 and normalizes pixels to [-1, 1]
<code>predict(model, labels, frame)</code>	Runs inference and returns class label + confidence score
<code>lock_screen()</code>	Calls ctypes.windll.user32.LockWorkStation() to lock Windows
<code>main()</code>	Main loop: reads webcam, runs prediction, manages 5s timer, triggers lock

05 Demo & Discussion

Live Demo

1. Run: `python main.py`

→ Webcam window opens

2. Person present

→ Green text: Person detected: 1.00

3. Leave the seat

→ Red countdown: Locking in 5s...4...3...

4. After 5 seconds

→ Windows screen locks automatically

Limitations

- Windows-only (ctypes API)
- May misclassify in extreme lighting
- Requires webcam access

Future Improvements

- Cross-platform support (macOS/Linux)
- Adjustable lock delay via config file
- Sound alert before locking
- Multi-person awareness

Thank You

Questions & Answers

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